In this ultrasound project, we aim to test whether a medical vision-language model such as Google's MedGemma can interpret ultrasound images reliably. According to the *MedGemma Technical Report*, MedGemma uses a medically adapted vision encoder called MedSigLIP. This encoder was trained using more than 33 million medical image-text pairs. Most of these were from areas including radiology, dermatology and ophthalmology and histopathology. However, ultrasound was not listed as a dedicated training modality, so MedGemma's ability to interpret ultrasound images still needs to be evaluated.

Wang et al. introduced SonoInstruct, a large ultrasound-specific instruction dataset, and used it to fine-tune a model called Qwen3-VL-2B-Sono. This showed that ultrasound-specific training can improve a general vision-language model's ultrasound understanding. In our project, however, we will initially evaluate MedGemma using zero-shot prompting. This means that we will ask MedGemma to classify ultrasound images without first training it on our BUSI and CAMUS images.

The article 'multimodal instruction dataset and benchmark for ultrasound understanding' states that increasing model size alone may not solve ultrasound-understanding problems. Models also need suitable ultrasound examples that connect written instructions with the relevant anatomy in the image. This is important because ultrasound images can be noisy, difficult to interpret and highly dependent on image quality.

Furthermore, if MedGemma is unable to precisely recognise an image, then it should be able to flag it as "insufficient image quality." Doing so, can help to prevent any diagnostic risks.

Based on this literature, our project will use fixed prompts and dataset splits, preserve every raw response and measure classification, localisation and failure to abstain. The results will be treated as a research evaluation rather than evidence that MedGemma is ready for clinical diagnosis.

Resources:

MedGemma Technical Report

A multimodal instruction dataset and benchmark for ultrasound understanding | npj Digital Medicine

https://pubmed.ncbi.nlm.nih.gov/42362710/

[2505.17779] U2-BENCH: Benchmarking Large Vision-Language Models on Ultrasound Understanding

Dataset of breast ultrasound images - ScienceDirect